

## packages/graph-core/src/analyzer/tour-generator.ts

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### Overview

A new file `packages/graph-core/src/analyzer/tour-generator.ts` adds a lightweight tour-generation system. It exports three functions and imports the `KnowledgeGraph` and `TourStep` types from `../types.js` (line 1). The functions are:

- `buildTourGenerationPrompt` (lines 7-62) – builds an LLM prompt that includes project metadata, node and edge lists, and layer information, and specifies a JSON response format.
- `parseTourGenerationResponse` (lines 70-120) – extracts JSON from raw or fenced text, validates required fields, and returns an array of `TourStep`.
- `generateHeuristicTour` (lines 135-293) – heuristically builds a tour using Kahn's topological sort, groups nodes by layers when present, batches code nodes into steps, and appends a final "Key Concepts" step for concept nodes.

### Key changes

- Import added: `KnowledgeGraph`, `TourStep` (line 1).
- Prompt construction, parsing, and heuristic generation functions added (lines 7-62, 70-120, 135-293).
- Documentation comments added for each function.
- Topological sort handles isolated nodes and cycles by appending them after the main pass (lines 181-186).
- Concept nodes are always appended last (lines 274-285).

### Impact

- **Algorithmic:** Uses Kahn's algorithm ( $O(V+E)$ ) for ordering code nodes.
- **Structure:** Centralizes prompt logic and parsing, reducing duplication.
- **Extensibility:** New functions are additive; existing exports remain unchanged.

### Risks & follow-ups

- `parseTourGenerationResponse` silently returns an empty array on any parsing error (lines 117-119); downstream code should handle empty tours.
- Layer ordering is derived from the topological sort; no evidence that layer dependencies are fully respected.
- Concept nodes are appended after all code nodes; if a concept node depends on code, the current ordering may mislead users.
- Prompt size grows with graph size; no evidence on token limits from the diff.